Part 3 | Mid-Semester Progress Report

Ensembles, regularization and hyperparameter optimisation: predicting student dropout and academic success

CS-13410 Introduction to Machine Learning · Fall 2026 · Assignment 3 | **Instructor:** Salman Ali
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Repository: github.com/maangatmaangat7-max/IML-7B-Project (notebook: *assignment 3/Part 3 - Ensembles and Hyperparameter Optimisation.ipynb*)

1. Work completed so far

- **Part 1 (done):** three-class problem (Dropout / Enrolled / Graduate) on the UCI dropout dataset (4,424 students, 36 predictors). Stratified 80/20 split, leakage-free *Pipeline/ColumnTransformer* (median imputation, train-fitted IQR clipping, robust scaling, one-hot encoding, two constructed approval-rate features), EDA.
- **Part 2 (done):** Naive Bayes vs. decision tree with nested cross-validation and a hand-worked information-gain/Gini split. Tuned tree and Bernoulli NB tied at macro F1 about 0.67; Gaussian NB on the full one-hot matrix failed (below the majority-class reference), with the cause diagnosed.
- **Part 3 (done, this report):** Random Forest, AdaBoost and **XGBoost**, plus Ridge / Lasso / Elastic-Net regularized logistic regression. Ensembles tuned by randomized search (15 draws per inner search, 40 for the final fit) and the regularized model by grid search; all tuned scores come from **nested CV** (5 folds × 2 repeats, tuning inside each training fold). Imbalance is handled with fold-local class/sample weights, never resampling. The test set was scored once.

2. Part 3 results

Model	Nested-CV macro F1 (10 folds)	CV ROC-AUC	CV recall Enrolled	Test macro F1 [95% CI]	Test ROC-AUC
Part 2: Bernoulli NB	0.671 ± 0.012	0.858	0.583	0.678 [0.646, 0.709]	0.860
Part 2: decision tree (tuned)	0.678 ± 0.018	0.847	0.561	0.674 [0.642, 0.704]	0.848
Regularized logistic (Lasso, C≈0.32)	0.720 ± 0.015	0.890	0.621	0.701 [0.668, 0.731]	0.886
Random Forest (tuned)	0.721 ± 0.012	0.890	0.610	0.721 [0.689, 0.751]	0.889
AdaBoost (tuned)	0.715 ± 0.018	0.884	0.658	0.685 [0.653, 0.714]	0.866
XGBoost (tuned)	**0.725 ± 0.012**	**0.896**	0.615	0.694 [0.661, 0.724]	**0.892**

- **Real improvement over Part 2.** XGBoost beats the Part 2 tree by +0.047 macro F1 (corrected paired p = 0.003) and +0.049 ROC-AUC (p < 0.001) in nested CV. XGBoost was chosen as the best model from nested-CV scores before looking at the test set.
- **The Part 3 models are statistically close to each other** on macro F1 (XGBoost vs Random Forest p = 0.22, vs logistic regression p = 0.40); XGBoost's clear edge is ROC-AUC. A Lasso-regularized *linear* model matches the ensembles using roughly 95 of 239 features, so the ceiling near 0.72-0.73 looks like a limit of the data rather than the model class.
- **Test set vs CV.** Random Forest leads on the 885-student test set (0.721) while XGBoost leads in CV; the bootstrap intervals (about ±0.03) overlap, so we treat this as noise and do not re-select on test. Tuning gave modest gains over defaults (+0.009 to +0.024 macro F1, not individually significant) but raised *Enrolled* recall for Random Forest from 0.35 to 0.61.
- **Features:** both ensembles rank the Part 1 feature *2nd-semester approval rate* first, then approved units and tuition status or grades; XGBoost also uses parental occupation/qualification codes.

3. Remaining plan

Part	Due	Plan
Part 4 (Assignment 4)	17-12-2026 (Week 15)	MLP in Keras or PyTorch on the same pipeline with dropout, weight decay and early stopping, compared against the best Part 3 model with the same nested/held-out protocol. Unsupervised step: k-means or hierarchical clustering and PCA on the prepared features, interpreted against the three outcomes. Interpretability and fairness: SHAP / permutation importance for the best model and a bias analysis across gender, nationality, marital status and the parental-background proxies.
Part 5 (final)	Week 16 (date TBA)	Single integrated notebook, GitHub repo cleanup, 8-12 page report (introduction, data, methods, results, discussion, conclusion, references), deployment/monitoring note, responsible-use statement, 10-12 minute presentation and individual viva preparation.

4. Risks and mitigations

- **Accuracy ceiling.** Four very different model families land at about 0.72 macro F1, so a neural network is unlikely to add much on 3,539 training rows. *Mitigation:* present Part 4 as a fair comparison (same protocol), accept a null result, and use the remaining effort on interpretability and fairness.
- **Weak minority class.** *Enrolled* precision is still only about 0.42. *Mitigation:* report per-class metrics everywhere; consider threshold tuning on cross-validated probabilities in Part 4.

- **Small test set / multiple comparisons.** 159 Enrolled test students make single-split differences of ±0.03 uninformative. *Mitigation:* keep decisions on nested CV and keep reporting bootstrap intervals.
- **Compute and tooling.** Nested tuning is expensive (the AdaBoost run alone took about 25 minutes on four cores) and deep-learning frameworks must be installed and tested on Windows. *Mitigation:* reuse saved search results where possible, fix random seeds, and install and smoke-test the framework before the Part 4 deadline.
- **Fairness and deployment.** Models use second-semester data, so they are end-of-year-1 early-warning tools, not enrollment-time predictors; the data come from one Portuguese institution and include sensitive attributes and proxies. *Mitigation:* state these limits explicitly in Parts 4-5.
- **Viva readiness.** Every member must be able to explain any part of the pipeline. *Mitigation:* rotate who reviews and explains each part before submission.